Comparative Analysis of Multi-Class Classifiers on Botanical Morphological Data

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Abstract:
The Fisher Iris dataset represents a foundational benchmark in multivariate pattern recognition. In this study, we

1. Morphological Dimensions:
Species discrimination within genus Iris (setosa, versicolor, virginica) relies on 4 measurements: sepal length, se

2. Experimental Methodology:
- Feature Preprocessing: Z-score normalization and median imputation.
- Data Partitioning: Stratified 80/20 train-test split, seed 42.
- Baseline: Logistic Regression (C=1.0, max_iter 1000).
- Proposed Model: Random Forest Classifier (100 estimators, Gini criterion).

3. Findings and Evidence:
1. Separability: Petal length and petal width account for >85% of tree feature importance.
2. Performance Comparison: On standardized test partitions, Random Forest achieved an overall accuracy of 96
3. Hyperparameter Sensitivity: Parameter sweeps on ensemble size (n in [10, 50, 100, 200]) revealed diminishin
4. Threats to Validity: Due to compact sample size of 150 specimens, sampling variance across test folds can sh

4. Conclusion:
Non-linear tree ensembles provide robust margin separation for overlapping botanical morphology, demonstratin